

MODULE 4 BRIDGES

4.1 INTRODUCTION

A bridge circuit in its simplest form consists of a network of four resistance arms forming a closed circuit, with a dc source of current applied to two opposite junctions and a current detector connected to the other two junctions, as shown in Figure 4.1.

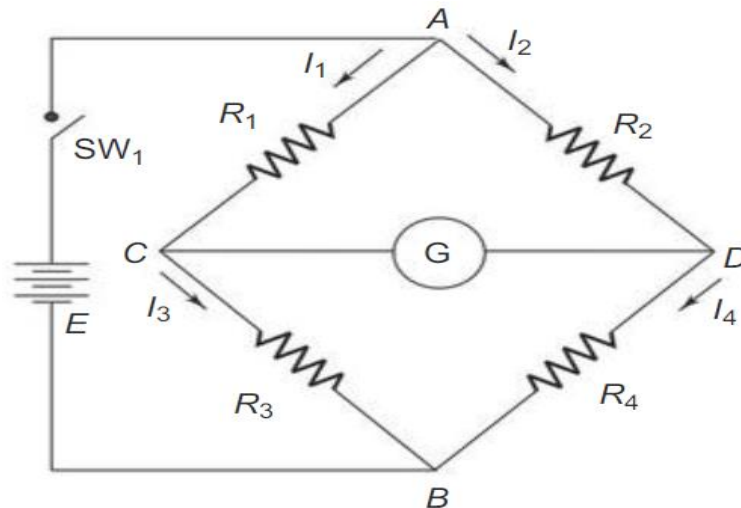


Figure 4.1 Wheatstone's bridge

Bridge circuits are extensively used for measuring component values such as R, L and C. Since the bridge circuit merely compares the value of an unknown component with that of an accurately known component (a standard), its measurement accuracy can be very high. This is because the readout of this comparison is based on the null indication at bridge balance and is essentially independent of the characteristics of the null detector. The measurement accuracy is therefore directly related to the accuracy of the bridge component and not to that of the null indicator used. The basic dc bridge is used for accurate measurement of resistance and is called Wheatstone's bridge.

4.2 WHEATSTONE BRIDGE (MEASUREMENT OF RESISTANCE)

Wheatstone's bridge is the most accurate method available for measuring resistances and is popular for laboratory use. The circuit diagram of a typical Wheatstone bridge is given in Figure 4.1. The source of emf and switch is connected to points A and B, while a sensitive current indicating meter, the galvanometer, is connected to points C and D. The galvanometer is a sensitive microammeter, with a zero center scale. When there is no current through the meter, the galvanometer pointer rests at 0, i.e. mid-scale. Current in one direction causes the pointer to deflect on one side and current in the opposite direction to the other side.

When SW1 is closed, current flows and divides into the two arms at point A, i.e. I_1 and I_2 . The bridge is balanced when there is no current through the galvanometer, or when the potential difference at points C and D is equal, i.e. the potential across the galvanometer is zero. To obtain the bridge balance equation, we have from the Figure 4.1.

$$I_1 R_1 = I_2 R_2 \quad \text{-----(4.1)}$$

For the galvanometer current to be zero, the following conditions should be satisfied.

$$I_1 = I_3 = \frac{E}{R_1 + R_3} \quad \text{-----(4.2)}$$

$$I_2 = I_4 = \frac{E}{R_2 + R_4} \text{ -----(4.3)}$$

Substituting in (4.1) we get,

$$\frac{E \times R_1}{R_1 + R_3} = \frac{E \times R_2}{R_2 + R_4}$$

$$R_1 \times (R_2 + R_4) = (R_1 + R_3) \times R_2$$

$$R_1 R_2 + R_1 R_4 = R_1 R_2 + R_3 R_2$$

$$R_4 = \frac{R_2 R_3}{R_1}$$

This is the equation for the bridge to be balanced. In a practical Wheatstone's bridge, at least one of the resistance is made adjustable, to permit balancing. When the bridge is balanced, the unknown resistance (normally connected at R_4) may be determined from the setting of the adjustable resistor, which is called a standard resistor because it is a precision device having very small tolerance.

Hence,

$$R_x = \frac{R_2 R_3}{R_1} \text{ -----(4.4)}$$

Example 1

Figure 4.1 consists of the following parameters. $R_1 = 10 \text{ k}\Omega$, $R_2 = 15 \text{ k}\Omega$ and $R_3 = 40 \text{ k}\Omega$. Find the unknown resistance R_x .

Solution:

From the equation for bridge balance we have,

$$R_1 R_4 = R_2 R_3, \text{ i.e. } R_1 R_x = R_2 R_3$$

Therefore,

$$R_x = \frac{R_2 R_3}{R_1} = \frac{15 \text{ k} \times 40 \text{ k}}{10 \text{ k}} = 60 \text{ k}\Omega$$

4.2.1 Sensitivity of a Wheatstone Bridge

When the bridge is in an unbalanced condition, current flows through the galvanometer, causing a deflection of its pointer. The amount of deflection is a function of the sensitivity of the galvanometer. Sensitivity can be thought of as deflection per unit current. A more sensitive galvanometer deflects by a greater amount for the same current. Deflection may be expressed in linear or angular units of measure, and sensitivity can be expressed in units of $S = \text{mm} / \mu\text{A}$ or Degree / μA or radians / μA .

Therefore it follows that the total deflection D is $D = S \times I$, where S is defined above and I is the current in microamperes.

4.2.2 Unbalanced Wheatstone's Bridge

To determine the amount of deflection that would result for a particular degree of unbalance, general circuit analysis can be applied, but we shall use Thevenin's theorem. Since we are interested in determining the current through the galvanometer, we wish to find the Thevenin's equivalent, as seen by the galvanometer. Thevenin's equivalent voltage is found by disconnecting the galvanometer from the bridge circuit, as shown in Figure 4.2, and determining the open circuit voltage between terminals a and b.

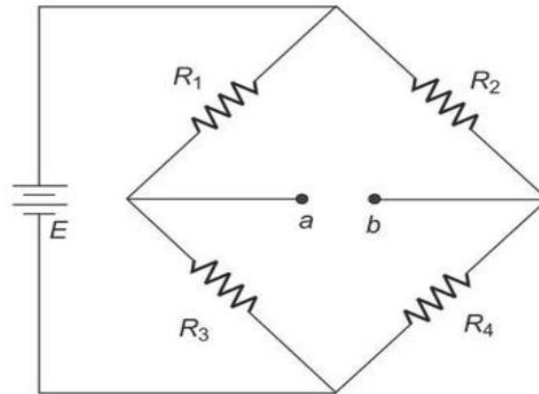


Figure 4.2 Unbalanced Wheatstone's Bridge

Applying the voltage divider equation, the voltage at point a can be determined as follows:

$$E_a = \frac{E \times R_3}{R_1 + R_3}$$

And at point b,

$$E_b = \frac{E \times R_4}{R_2 + R_4}$$

Therefore, the voltage between a and b is the difference between E_a and E_b , which represents Thevenin's equivalent voltage.

$$E_{th} = E_{ab} = E_a - E_b = \frac{E \times R_3}{R_1 + R_3} - \frac{E \times R_4}{R_2 + R_4}$$

Therefore,

$$E_{ab} = E \left(\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right)$$

Thevenin's equivalent resistance can be determined by replacing the voltage source E with its internal impedance or otherwise short-circuited and calculating the resistance looking into terminals a and b. Since the internal resistance is assumed to be very low, we treat it as 0Ω . Thevenin's equivalent circuit is shown in Figure 4.3.

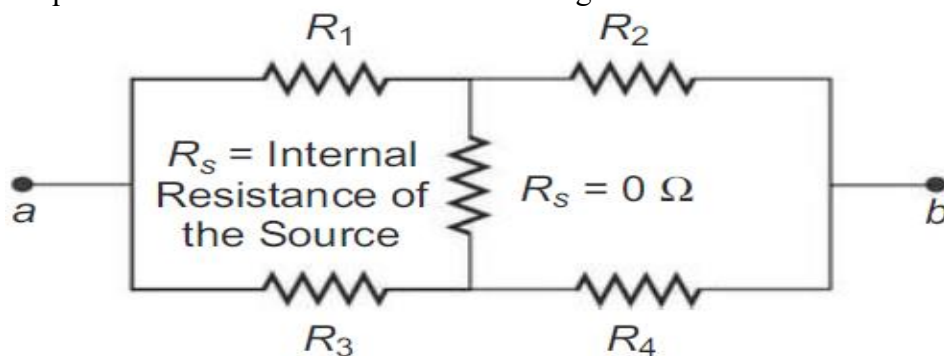


Figure 4.3 Thevenin's resistance

The equivalent resistance of the circuit is $R_1 // R_3$ in series with $R_2 // R_4$ i.e. $R_1 // R_3 + R_2 // R_4$.

$$R_{th} = \frac{R_1 R_3}{R_1 + R_3} + \frac{R_2 R_4}{R_2 + R_4}$$

Therefore, Thevenin's equivalent circuit is given in Figure 4.4. Thevenin's equivalent circuit for the bridge, as seen looking back at terminals a and b in Figure 4.2, is shown in Figure 4.4. If a galvanometer is connected across the terminals a and b of Figure 4.2, or its Thevenin equivalent Figure 4.4 it will experience the same deflection at the output of the bridge. The magnitude of current is limited by both Thevenin's equivalent resistance and any resistance connected between a and b. The resistance between a and b consists only of the galvanometer resistance R_g . The deflection current in the galvanometer is therefore given by

$$I_g = \frac{E_{th}}{R_{th} + R_g} \text{-----(4.5)}$$

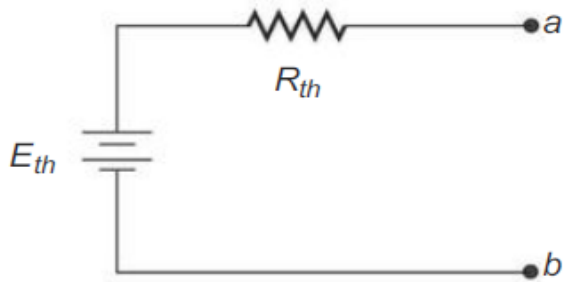
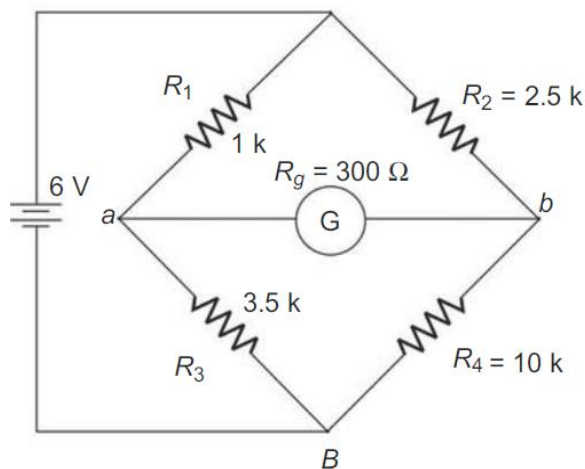


Figure 4.4 Thevenin's equivalent

Example 2

An unbalanced Wheatstone bridge is given in Figure. Calculate the current through the galvanometer.

Solution:



The Thevenin's equivalent voltage between a and b is the difference of voltages at these points.
 $E_{th} = E_a - E_b = E_b - E_a$.

$$\therefore E_{th} = E \left(\frac{R_4}{R_2 + R_4} - \frac{R_3}{R_1 + R_3} \right)$$

$$E_{th} = 6 \left(\frac{10 \text{ k}}{2.5 \text{ k} + 10 \text{ k}} - \frac{3.5 \text{ k}}{1 \text{ k} + 3.5 \text{ k}} \right)$$

$$E_{th} = 6 (0.800 - 0.778)$$

$$E_{th} = 0.132 \text{ V}$$

Thevenin's equivalent resistance is

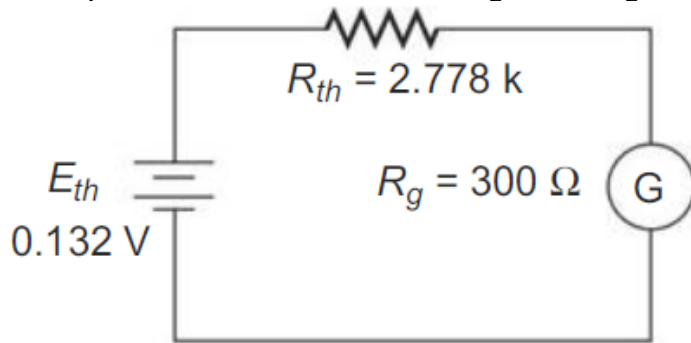
$$R_{th} = \frac{R_1 R_3}{R_1 + R_3} + \frac{R_2 R_4}{R_2 + R_4}$$

$$R_{th} = \frac{1 \text{ k} \times 3.5 \text{ k}}{1 \text{ k} + 3.5 \text{ k}} + \frac{2.5 \text{ k} \times 10 \text{ k}}{2.5 \text{ k} + 10 \text{ k}}$$

$$= 0.778 \text{ k} + 2 \text{ k}$$

$$= 2.778 \text{ k}$$

The equivalent circuit connected along with the galvanometer is as shown.



Equivalent circuit

The current through the galvanometer is given by,

$$I_g = \frac{E_{th}}{R_{th} + R_g} = \frac{0.132 \text{ V}}{2.778 \text{ k} + 0.3 \text{ k}} = 42.88 \mu\text{A}$$

4.2.3 Slightly Unbalanced Wheatstone's Bridge

If three of the four resistors in a bridge are equal to R and the fourth differs by 5% or less, we can develop an approximate but accurate expression for Thevenin's equivalent voltage and resistance.

Consider the circuit in Figure 4.5.

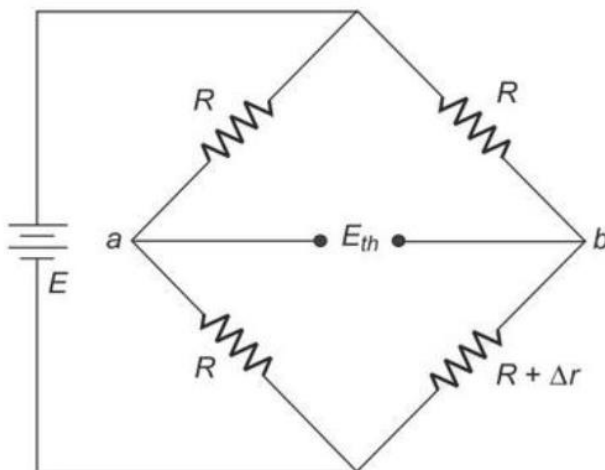


Figure 4.5 Slightly unbalanced Wheatstone's bridge

The voltage at point a is

$$E_a = \frac{E \times R}{R + R} = \frac{E \times R}{2R} = \frac{E}{2}$$

The voltage at point b is

$$E_b = \frac{R + \Delta r \times E}{R + R + \Delta r} = \frac{E(R + \Delta r)}{2R + \Delta r}$$

Thevenin's equivalent voltage between a and b is the difference between these voltages.

Therefore

$$\begin{aligned} E_{th} &= E_a - E_b = E \left(\frac{(R + \Delta r)}{2R + \Delta r} - \frac{1}{2} \right) \\ &= E \left(\frac{2(R + \Delta r) - (2R + \Delta r)}{2(2R + \Delta r)} \right) \\ &= E \left(\frac{2R + 2\Delta r - 2R - \Delta r}{4R + 2\Delta r} \right) \\ &= E \left(\frac{\Delta r}{4R + 2\Delta r} \right) \end{aligned}$$

If Δr is 5% of R or less, Δr in the denominator can be neglected without introducing appreciable error. Therefore, Thevenin's voltage is

$$E_{th} = \frac{E \times \Delta r}{4R} = E \left(\frac{\Delta r}{4R} \right)$$

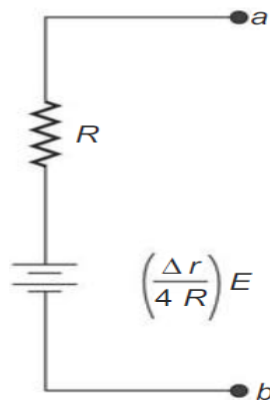


Figure 4.6 Thevenin's equivalent of a slightly unbalanced Wheatstone's bridge

The equivalent resistance can be calculated by replacing the voltage source with its internal impedance (for all practical purpose short-circuit). The Thevenin's equivalent resistance is given by

$$\begin{aligned} R_{th} &= \frac{R \times R}{R + R} + \frac{R(R + \Delta r)}{R + R + \Delta r} \\ &= \frac{R}{2} + \frac{R(R + \Delta r)}{2R + \Delta r} \end{aligned}$$

Again, if Δr is small compared to R , Δr can be neglected. Therefore,

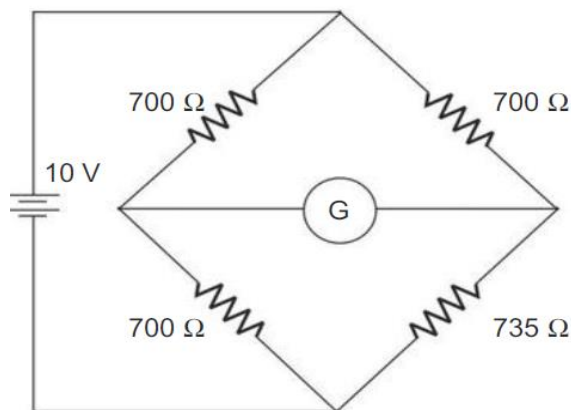
$$R_{th} = \frac{R}{2} + \frac{R}{2} = R$$

Using these approximations, the Thevenin's equivalent circuit is as shown in Figure 4.6. These approximate equations are about 98% accurate if:

$$\Delta r \leq 0.05 R.$$

Example 3

Given a centre zero 200 – 0 – 200 mA movement having an internal resistance of 125 Ω . Calculate the current through the galvanometer given in Figure by the approximation method.



Solution:

The Thevenin's equivalent voltage is

$$E_{th} = \frac{E (\Delta r)}{4 R}$$

$$= \frac{10 \times 35}{4 \times 700} = 0.125 \text{ V}$$

Thevenin's equivalent resistance is $R_{th} = R = 700 \Omega$

The current through the galvanometer is

$$I_g = \frac{E_{th}}{R_{th} + R_g} = \frac{0.125 \text{ V}}{700 + 125} = \frac{0.125}{825} = 151.5 \mu\text{A}$$

If the detector is a 200 – 0 – 200 μA galvanometer, we see that the pointer is full scale for a 5% change in resistance.

4.2.4 Application of Wheatstone's Bridge

A Wheatstone bridge may be used to measure the dc resistance of various types of wire, either for the purpose of quality control of the wire itself, or of some assembly in which it is used. For example, the resistance of motor windings, transformers, solenoids, and relay coils can be measured. Wheatstone's bridge is also used extensively by telephone companies and others to locate cable faults. The fault may be two lines shorted together, or a single line shorted to ground.

4-2.5 Limitations of Wheatstone's Bridge

For low resistance measurement, the resistance of the leads and contacts becomes significant and introduces an error. This can be eliminated by Kelvin's Double bridge. For high resistance measurements, the resistance presented by the bridge becomes so large that the galvanometer is insensitive to imbalance. Therefore, a power supply has to replace the battery and a dc VTVM replaces the galvanometer. In the case of high resistance measurements in mega ohms, the Wheatstone bridge cannot be used. Another difficulty in Wheatstone's bridge is the change in resistance of the bridge arms due to the heating effect of current through the resistance. The rise in temperature causes a change in the value of the resistance, and excessive current may cause a permanent change in value.

4.3 KELVIN'S BRIDGE

When the resistance to be measured is of the order of magnitude of bridge contact and lead resistance, a modified form of Wheatstone's bridge, the Kelvin bridge is employed. Kelvin's bridge is a modification of Wheatstone's bridge and is used to measure values of resistance below 1Ω . In low resistance measurement, the resistance of the leads connecting the unknown resistance to the terminal of the bridge circuit may affect the measurement.

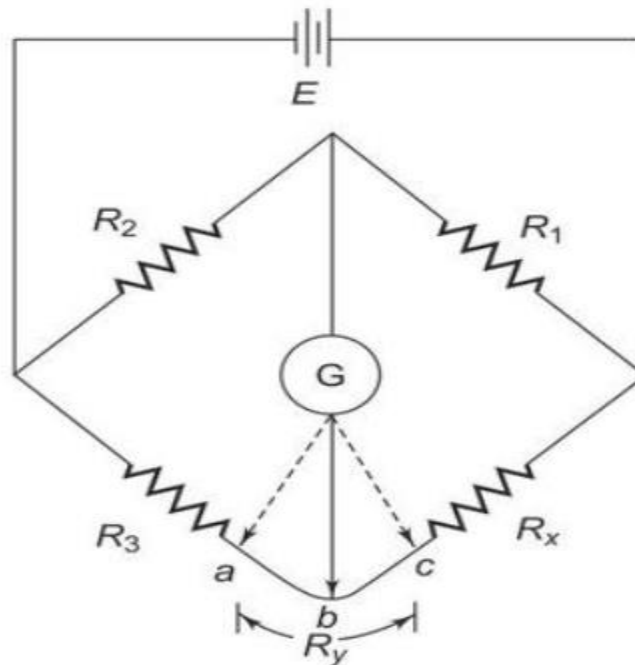


Figure 4.7 Kelvin's bridge

Consider the circuit in Figure 4.7, where R_y represents the resistance of the connecting leads from R_3 to R_x (unknown resistance). The galvanometer can be connected either to point c or to point a. When it is connected to point a, the resistance R_y , of the connecting lead is added to the unknown resistance R_x , resulting in too high indication for R_x . When the connection is made to point c, R_y is added to the bridge arm R_3 and resulting measurement of R_x is lower than the actual value, because now the actual value of R_3 is higher than its nominal value by the resistance R_y . If the galvanometer is connected to point b, in between points c and a, in such a way that the ratio of the resistance from c to b and that from a to b equals the ratio of resistances R_1 and R_2 , then

$$\frac{R_{cb}}{R_{ab}} = \frac{R_1}{R_2} \text{ -----(4.6)}$$

and the usual balance equations for the bridge give the relationship

$$(R_x + R_{cb}) = \frac{R_1}{R_2} (R_3 + R_{ab}) \text{-----(4.7)}$$

But

$$R_{ab} + R_{cb} = R_y \text{ and } \frac{R_{cb}}{R_{ab}} = \frac{R_1}{R_2}$$

$$\frac{R_{cb}}{R_{ab}} + 1 = \frac{R_1}{R_2} + 1$$

$$\frac{R_{cb} + R_{ab}}{R_{ab}} = \frac{R_1 + R_2}{R_2}$$

$$\frac{R_y}{R_{ab}} = \frac{R_1 + R_2}{R_2}$$

Therefore

$$R_{ab} = \frac{R_2 R_y}{R_1 + R_2}$$

and as

$$R_{ab} + R_{cb} = R_y$$

$$R_{cb} = R_y - R_{ab} = R_y - \frac{R_2 R_y}{R_1 + R_2}$$

$$R_{cb} = \frac{R_1 R_y + R_2 R_y - R_2 R_y}{R_1 + R_2} = \frac{R_1 R_y}{R_1 + R_2}$$

Substituting for R_{ab} and R_{cb} in Equation (4.7), we have,

$$R_x + \frac{R_1 R_y}{R_1 + R_2} = \frac{R_1}{R_2} \left(R_3 + \frac{R_2 R_y}{R_1 + R_2} \right)$$

$$R_x + \frac{R_1 R_y}{R_1 + R_2} = \frac{R_1 R_3}{R_2} + \frac{R_1 R_2 R_y}{R_2 (R_1 + R_2)}$$

Hence

$$R_x = \frac{R_1 R_3}{R_2} \text{-----(4.8)}$$

Equation (4.8) is the usual Wheatstone's balance equation and it indicates that the effect of the resistance of the connecting leads from point a to point c has been eliminated by connecting the galvanometer to an intermediate position, b.

AC BRIDGES

Impedances at AF or RF are commonly determined by means of an ac Wheatstone bridge. The diagram of an ac bridge is given in Figure 4.8. This bridge is similar to a dc bridge, except that the bridge arms are impedances. The bridge is excited by an ac source rather than dc and the galvanometer is replaced by a detector, such as a pair of headphones, for detecting AC.

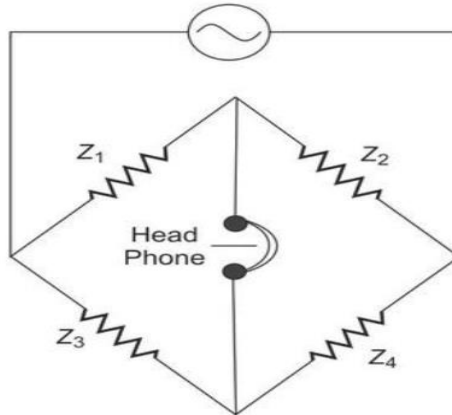


Figure 4.8 AC Wheatstone's bridge

When the bridge is balanced,

$$\frac{Z_1}{Z_3} = \frac{Z_2}{Z_4}$$

where Z_1 , Z_2 , Z_3 and Z_4 are the impedances of the arms, and are vector complex quantities that possess phase angles. It is thus necessary to adjust both the magnitude and phase angles of the impedance arms to achieve balance, i.e. the bridge must be balanced for both the reactance and the resistive component.

4.4 CAPACITANCE COMPARISON BRIDGE

Figure 4.9 shows the circuit of a capacitance comparison bridge. The ratio arms R_1 , R_2 are resistive. The known standard capacitor C_3 is in series with R_3 . R_3 may also include an added variable resistance needed to balance the bridge. C_x is the unknown capacitor and R_x is the small leakage resistance of the capacitor. In this case an unknown capacitor is compared with a standard capacitor and the value of the former, along with its leakage resistance, is obtained.

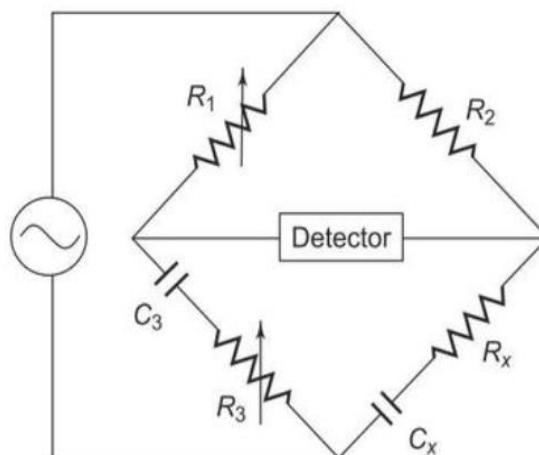


Figure 4.9 Capacitance comparison bridge

Hence,

$$Z_1 = R_1$$

$$Z_2 = R_2$$

$$Z_3 = R_3 \text{ in series with } C_3 = R_3 - j/\omega C_3$$

$$Z_x = R_x \text{ in series with } C_x = R_x - j/\omega C_x$$

The condition for balance of the bridge is

$$Z_1 Z_x = Z_2 Z_3$$

$$R_1 \left(R_x - \frac{j}{\omega C_x} \right) = R_2 \left(R_3 - \frac{j}{\omega C_3} \right)$$

$$R_1 R_x - \frac{j R_1}{\omega C_x} = R_2 R_3 - \frac{j R_2}{\omega C_3}$$

Two complex quantities are equal when both their real and their imaginary terms are equal.

Therefore,

$$R_1 R_x = R_2 R_3$$

$$\therefore R_x = \frac{R_2 R_3}{R_1} \text{ -----[4.9(a)]}$$

$$\frac{R_1}{\omega C_x} = \frac{R_2}{\omega C_3}$$

$$C_x = \frac{C_3 R_1}{R_2} \text{ -----[4.9(b)]}$$

Example 1

A capacitance comparison bridge (similar angle bridge) is used to measure a capacitive impedance at a frequency of 2 kHz. The bridge constants at balance are $C_3 = 100 \text{ mF}$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 50 \text{ k}\Omega$, $R_3 = 100 \text{ k}\Omega$. Find the equivalent series circuit of the unknown impedance.

Solution:

Finding R_x using the equation

$$R_x = \frac{R_2 R_3}{R_1}$$

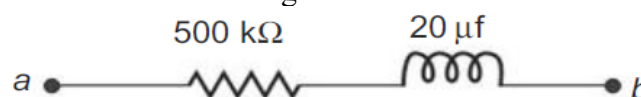
$$= \frac{100 \text{ k} \times 50 \text{ k}}{10 \text{ k}} = 500 \text{ k}\Omega$$

Then finding C_x using the equation,

$$C_x = \frac{R_1}{R_2} C_3$$

$$= \frac{10 \text{ k}}{50 \text{ k}} \times 100 \times 10^{-6} = 20 \text{ }\mu\text{F}$$

The equivalent series circuit is shown in Figure.



4.5 INDUCTANCE COMPARISON BRIDGE

Figure 4.10 gives a schematic diagram of an inductance comparison bridge. In this, values of the unknown inductance L_x and its internal resistance R_x are obtained by comparison with the standard inductor and resistance, i.e. L_3 and R_3 .

The equation for balance condition is

$$Z_1 Z_x = Z_2 Z_3$$

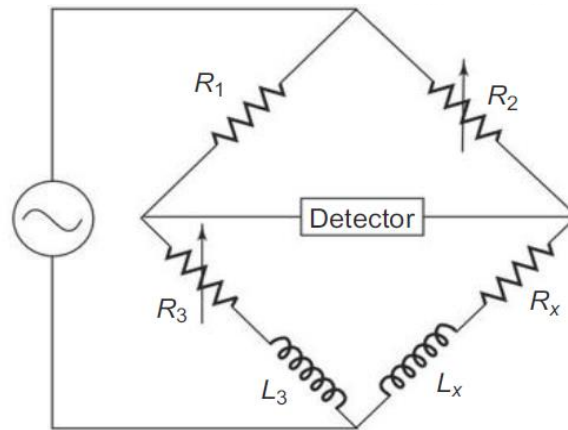


Figure 4.10 Inductance comparison bridge

Hence,

$$Z_1 = R_1$$

$$Z_2 = R_2$$

$$Z_3 = R_3 \text{ in series with } L_3 = R_3 + j \omega L_3$$

$$Z_x = R_x \text{ in series with } L_x = R_x + j \omega L_x$$

The condition for balance of the bridge is

$$Z_1 Z_x = Z_2 Z_3$$

$$R_1 (R_x + j \omega L_x) = R_2 (R_3 + j \omega L_3)$$

$$R_1 R_x + j R_1 \omega L_x = R_2 R_3 + j R_2 \omega L_3$$

Two complex quantities are equal when both their real and their imaginary terms are equal.

Therefore,

$$R_1 R_x = R_2 R_3$$

$$\therefore R_x = \frac{R_2 R_3}{R_1} \text{ -----[4.10(a)]}$$

$$R_1 \omega L_x = R_2 \omega L_3$$

$$L_x = \frac{L_3 R_2}{R_1} \text{ -----[4.10(b)]}$$

In this bridge R_2 is chosen as the inductive balance control and R_3 as the resistance balance control. (It is advisable to use a fixed resistance ratio and variable standards). Balance is obtained by alternately varying L_3 or R_3 . If the Q of the unknown reactance is greater than the standard Q , it is necessary to place a variable resistance in series with the unknown reactance to obtain balance. If the unknown inductance has a high Q , it is permissible to vary the resistance ratio when a variable standard inductor is not available.

Example 1

An inductance comparison bridge is used to measure inductive impedance at a frequency of 5 KHz. The bridge constants at balance are $L_3 = 10 \text{ mH}$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 40 \text{ k}\Omega$, $R_3 = 100 \text{ k}\Omega$. Find the equivalent series circuit of the unknown impedance.

Solution:

Given $L_3 = 10 \text{ mH}$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 40 \text{ k}\Omega$, $R_3 = 100 \text{ k}\Omega$. To find R_x and L_x

From balance equation,

$$R_x = \frac{R_2 R_3}{R_1} = \frac{40 \text{ K} \times 100 \text{ K}}{10 \text{ K}} = 400 \text{ k}\Omega$$

$$L_x = \frac{R_2 L_3}{R_1} = \frac{10 \text{ mH} \times 40 \text{ K}}{10 \text{ K}} = 40 \text{ mH}$$

The equivalent series circuit is shown in Figure.

**4.6 MAXWELL'S BRIDGE**

Maxwell's bridge, shown in Figure 4.11, measures an unknown inductance in terms of a known capacitor. The use of standard arm offers the advantage of compactness and easy shielding. The capacitor is almost a loss-less component. One arm has a resistance R_1 in parallel with C_1 , and hence it is easier to write the balance equation using the admittance of arm 1 instead of the impedance.

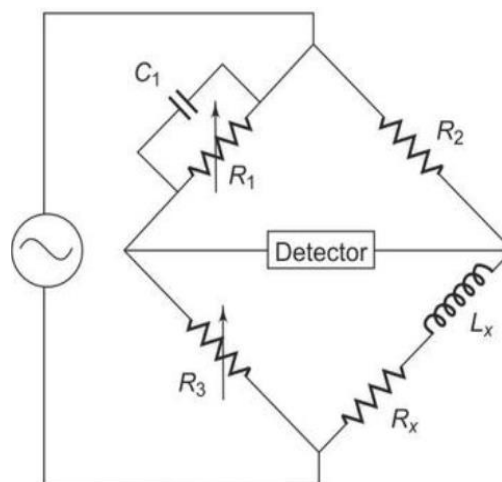


Figure 4.11 Maxwell's bridge

The general equation for bridge balance is,

$$Z_1 Z_x = Z_2 Z_3$$

$$Z_x = \frac{Z_2 Z_3}{Z_1} = Z_2 Z_3 Y_1 \quad \text{-----(4.11)}$$

where

$$Z_1 = R_1 \text{ in parallel with } C_1 \text{ i.e. } Y_1 = \frac{1}{Z_1}$$

$$Y_1 = \frac{1}{R_1} + j\omega C_1$$

$$Z_2 = R_2$$

$$Z_3 = R_3$$

$$Z_x = R_x \text{ in series with } L_x = R_x + j\omega L_x$$

From Eq. (4.14) we have,

$$R_x + j\omega L_x = R_2 R_3 \left(\frac{1}{R_1} + j\omega C_1 \right)$$

$$R_x + j\omega L_x = \frac{R_2 R_3}{R_1} + j\omega C_1 R_2 R_3$$

Equating real terms and imaginary terms we have,

$$R_x = \frac{R_2 R_3}{R_1}$$

$$L_x = C_1 R_2 R_3$$

Also,

$$Q = \frac{\omega L_x}{R_x} = \frac{\omega C_1 R_2 R_3 \times R_1}{R_2 R_3} = \omega C_1 R_1$$

Maxwell's bridge is limited to the measurement of low Q values (1 – 10). The measurement is independent of the excitation frequency. The scale of the resistance can be calibrated to read inductance directly. The Maxwell bridge using a fixed capacitor has the disadvantage that there is an interaction between the resistance and reactance balances. This can be avoided by varying the capacitances, instead of R_2 and R_3 , to obtain a reactance balance. However, the bridge can be made to read directly in Q.

The bridge is particularly suited for inductances measurements, since comparison with a capacitor is more ideal than with another inductance. Commercial bridges measure from 1 – 1000 H, with $\pm 2\%$ error. (If the Q is very large, R_1 becomes excessively large and it is impractical to obtain a satisfactory variable standard resistance in the range of values required).

Example 1

A Maxwell bridge is used to measure an inductive impedance. The bridge constants at balance are $C_1 = 0.01 \mu\text{F}$, $R_1 = 470 \text{ k}\Omega$, $R_2 = 5.1 \text{ k}\Omega$, and $R_3 = 100 \text{ k}\Omega$. Find the series equivalent of the unknown impedance.

Solution:

We need to find R_x and L_x

$$R_x = \frac{R_2 R_3}{R_1} = \frac{100 \text{ k} \times 5.1 \text{ k}}{470 \text{ k}} = 1.09 \text{ k}\Omega$$

$$\begin{aligned} L_x &= R_2 R_3 C_1 \\ &= 5.1 \text{ k} \times 100 \text{ k} \times 0.01 \mu\text{f} \\ &= 5.1 \text{ H} \end{aligned}$$

The equivalent series circuit is shown in Figure.



4.7 WIEN'S BRIDGE

The Wien bridge shown in Figure 4.12 has a series RC combination in one arm and a parallel combination in the adjoining arm. Wien's bridge in its basic form, is designed to measure frequency. It can also be used for the measurement of an unknown capacitor with great accuracy.

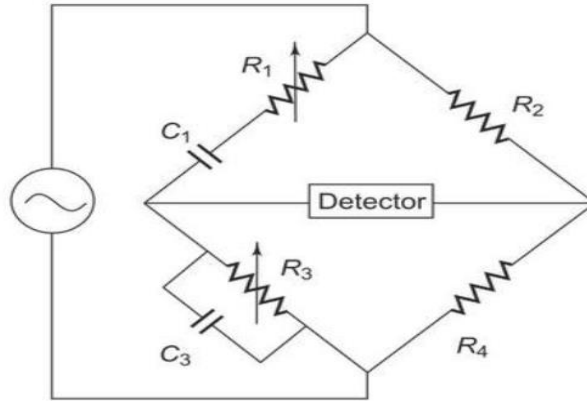


Figure 4.12 Wein's Bridge

The impedance of one arm is,

$$Z_1 = R_1 - j/\omega C_1$$

The admittance of the parallel arm is,

$$Y_3 = 1/R_3 + j \omega C_3$$

Using the bridge balance equation, we have,

$$Z_1 Z_4 = Z_2 Z_3$$

Therefore,

$$Z_1 Z_4 = Z_2/Y_3, \text{ i.e. } Z_2 = Z_1 Z_4 Y_3$$

$$R_2 = R_4 \left(R_1 - \frac{j}{\omega C_1} \right) \left(\frac{1}{R_3} + j \omega C_3 \right)$$

$$R_2 = \frac{R_1 R_4}{R_3} - \frac{j R_4}{\omega C_1 R_3} + j \omega C_3 R_1 R_4 + \frac{C_3 R_4}{C_1}$$

$$R_2 = \left(\frac{R_1 R_4}{R_3} + \frac{C_3 R_4}{C_1} \right) - j \left(\frac{R_4}{\omega C_1 R_3} - \omega C_3 R_1 R_4 \right)$$

Equating the real and imaginary terms we have,

$$R_2 = \frac{R_1 R_4}{R_3} + \frac{C_3 R_4}{C_1}$$

$$\frac{R_4}{\omega C_1 R_3} - \omega C_3 R_1 R_4 = 0$$

Therefore,

$$\frac{R_2}{R_4} = \frac{R_1}{R_3} + \frac{C_3}{C_1} \text{ -----(4.12)}$$

$$\frac{1}{\omega C_1 R_3} = \omega C_3 R_1 \text{-----(4.13)}$$

$$\omega^2 = \frac{1}{C_1 R_1 R_3 C_3}$$

$$\omega = \frac{1}{\sqrt{C_1 R_1 C_3 R_3}}$$

$$\omega = 2 \pi f$$

$$f = \frac{1}{2 \pi \sqrt{C_1 R_1 C_3 R_3}} \text{-----(4.14)}$$

The two conditions for bridge balance, (4.12) and (4.14), result in an expression determining the required resistance ratio R_2/R_4 and another expression determining the frequency of the applied voltage. If we satisfy Equation (4.12) and also excite the bridge with the frequency of Equation (4.14), the bridge will be balanced.

In most Wien bridge circuits, the components are chosen such that $R_1 = R_3 = R$ and $C_1 = C_3 = C$. Equation (4.12) therefore reduces to $R_2/R_4 = 2$ and Equation (4.14) to $f = 1/2\pi RC$, which is the general equation for the frequency of the bridge circuit.

The bridge is used for measuring frequency in the audio range. Resistances R_1 and R_3 can be ganged together to have identical values. Capacitors C_1 and C_3 are normally of fixed values. The audio range is normally divided into 20 – 200 – 2 k – 20 kHz ranges. In this case, the resistances can be used for range changing and capacitors C_1 and C_3 for fine frequency control within the range. The bridge can also be used for measuring capacitances. In that case, the frequency of operation must be known.

The bridge is also used in a harmonic distortion analyzer, as a Notch filter, and in audio frequency and radio frequency oscillators as a frequency determining element. An accuracy of 0.5% – 1% can be readily obtained using this bridge. Because it is frequency sensitive, it is difficult to balance unless the waveform of the applied voltage is purely sinusoidal.

Example 1

A Wein Bridge circuit consists of the following:

$R_1 = 4.7 \text{ k}\Omega$, $C_1 = 5 \text{ nf}$, $R_2 = 20 \text{ k}\Omega$, $C_3 = 10 \text{ nf}$, $R_3 = 10 \text{ k}\Omega$, $R_4 = 100 \text{ k}\Omega$

Determine the frequency of the circuit.

Solution:

The frequency is given by the equation,

$$f = \frac{1}{2 \pi \sqrt{C_1 R_1 R_3 C_3}}$$

$$f = \frac{1}{2 \pi \sqrt{5 \times 10^{-9} \times 4.7 \times 10^3 \times 10 \times 10^{-9} \times 10 \times 10^3}}$$

$$f = \frac{1}{2 \pi \sqrt{5 \times 10^{-10} \times 4.7}}$$

$$f = \frac{10^5}{2 \pi \sqrt{5 \times 4.7}} = 3.283 \text{ kHz}$$

Example 2

Find the equivalent parallel resistance and capacitance that causes a Wien bridge to null with the following component values.

$R_1 = 3.1 \text{ k}\Omega$, $C_1 = 5.2 \text{ }\mu\text{F}$, $R_2 = 25 \text{ k}\Omega$, $f = 2.5 \text{ kHz}$, $R_4 = 100 \text{ k}\Omega$

Solution:

Given $\omega = 2\pi f = 2 \times 3.14 \times 2500 = 15.71 \text{ k rad/s}$.

Substituting the value of C_3 from Equation (4.13) in Equation (4.12) we get,

$$\begin{aligned} R_3 &= \frac{R_4}{R_2} \left(R_1 + \frac{1}{\omega^2 R_1 C_1^2} \right) \\ &= \frac{100 \text{ k}}{25 \text{ k}} \left(3.1 \text{ k} + \frac{1}{(15.71 \text{ k})^2 \times 3.1 \text{ k} \times (5.2 \times 10^{-6})^2} \right) \\ &= 12.4 \text{ k}\Omega \\ C_3 &= \frac{R_2}{R_4} \left(\frac{C_1}{1 + \omega^2 R_1^2 C_1^2} \right) \\ &= \frac{25 \text{ k}}{100 \text{ k}} \left(\frac{5.2 \times 10^{-6}}{1 + (15.71 \text{ k})^2 \times (3.1 \text{ k})^2 \times (5.2 \times 10^{-6})^2} \right) \\ &= 1.3 \times 10^{-6} \left(\frac{1}{1 + 64133.07} \right) \\ &= 20.3 \text{ pf} \end{aligned}$$

The value of C_3 can also be found out by using equation,

$$C_3 = \frac{1}{\omega^2 C_1 R_1 R_3}$$

Example 3

An ac bridge with terminals ABCD has in

Arm AB a resistance of $800 \text{ }\Omega$ in parallel with a capacitor of $0.5 \text{ }\mu\text{F}$,

Arm BC – a resistance of $400 \text{ }\Omega$ in series with a capacitor of $1 \text{ }\mu\text{F}$,

Arm CD – a resistance of $1000 \text{ }\Omega$, Arm DA – a pure resistance R .

(a) Determine the value of frequency for which the bridge is balanced

(b) Calculate the value of R required to produce balance.

Solution:

The bridge configuration is of Wien Bridge.

Given: $C_1 = 0.5 \text{ }\mu\text{F}$, $R_1 = 800 \text{ }\Omega$, $C_2 = 1.0 \text{ }\mu\text{F}$, $R_2 = 400 \text{ }\Omega$, $R_4 = 1000 \text{ }\Omega$, $R_3 = R = ?$

$$\begin{aligned} f &= \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}} \\ &= \frac{1}{2\pi\sqrt{800 \times 0.5 \text{ }\mu\text{F} \times 400 \times 1 \text{ }\mu\text{F}}} \\ &= \frac{1}{2\pi\sqrt{800 \times 400 \times 0.5 \times 10^{-12}}} \\ &= \frac{10^6}{2\pi\sqrt{800 \times 200}} \\ &= \frac{10^6}{2\pi \times 400} = \frac{1000 \text{ kHz}}{2 \times 3.14 \times 400} = \frac{1000}{314 \times 8} = 0.398 \text{ kHz} \end{aligned}$$

Also, given

$$\frac{R_2}{R_1} + \frac{C_1}{C_2} = \frac{R_4}{R_3}$$
$$\frac{400}{800} + \frac{0.5 \mu\text{F}}{1 \mu\text{F}} = \frac{1000}{R}$$
$$0.5 + 0.5 = \frac{1000}{R}$$
$$R = 1000 \Omega$$